MachinePlan-10K: A BRep-Based Synthetic Dataset for Machining Process Planning with Tool-Aware Feature Sampling

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We present a dataset developed for machining process planning. The dataset was generated using Siemens Designcenter NX and contains 10,000 unique, validated parts as BRep geometries, along with wireframe and shaded CAD images, associated process plans, in-process workpiece (IPW) geometries, tool information, machining times, and tool paths. The dataset focuses on planar machining operations under 2.5-axis CNC machining and is intended for research and development related to machining process planning. All dimensions are in metric units (mm), and only features that can be machined using tools available in the NX CAM tool library are included. The automated data generation pipeline ensures consistent and accurate labeling, minimizing inter-annotator variability and discrepancies arising from manual annotation. Similarly, leveraging the NX knowledge base ensures that process planning rules are applied consistently in a deterministic way, eliminating variability from subjective judgments. The dataset is publicly available at https://doi.org/10.5281/zenodo.21653081.

Additional Key Words and Phrases: machining process planning, synthetic dataset, BRep, CNC machining, CAM, tool paths, machining sequence

1 Introduction

Machining process planning converts raw stock to a finished part via a sequence of machining operations. Process planning is difficult because each operation changes the geometry of the in-process workpiece (IPW), which in turn constrains the next feasible operations, tools, and parameters. Traditional workflows depend on expert CAM programming and are labor-intensive. This challenge is magnified in 2.5-axis planar milling, where pockets, slots, and holes must be sequenced and tooled under material, machine, and fixturing constraints. Recent research frames machining process planning as a learning problem using STL IPW snapshots, BRep design geometry, and textual process records to predict operation sequences and CAM details [5].

To facilitate the development of AI-based models for machining process planning, we present a dataset of 10,000 parts with a validated process plan and IPW snapshots. The ground-truth operations and machining sequences are simulated and validated using Siemens Designcenter NX (hereafter NX). The dataset includes a variety of features such as pockets, holes, and chamfers, with different dimensions and relationships among features. The features are distributed across the raw stock in a manner that ensures a variety of different machining scenarios.

2 Salient Features of the MachinePlan-10K Dataset

In this section, we highlight the key features of the MachinePlan-10K dataset that distinguish it from other datasets in the field of machining process planning.

The first is how the IPW itself is obtained. We create it using the NX CAM module, which simulates the machining operations and generates the IPW geometry after each operation. Unlike other datasets that rely on CAD subtraction, this simulation-based approach accounts for tool geometry, tool orientation, and tool path, so that the IPW is a realistic

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representation of the part after each operation rather than a theoretical subtraction of the feature from the raw stock. Each IPW is exported as a mesh in STL format, which can be used for training models to predict the next operation from the current state of the part.

The features themselves are likewise generated with the availability and limitations of tools in 2.5-axis CNC machining in mind. The depth of a pocket, for example, is limited by the flute length of the endmill used to machine it. Because a circular cutting tool cannot machine a sharp 90-degree corner, pocket corners need to be blended, and the resulting corner radius is in turn limited by the radius of the endmill available to machine it.

Beyond the geometry, we document which tools are used for each feature and operation. Tools matter because the same feature can be machined by different ones, and the choice of tool affects the machining time. A hole, for instance, can be drilled in a single operation using a drill, in multiple operations using a spot drill and a drill, or in multiple operations using a spot drill, a drill, and a reamer. A deep through-hole might need a pilot hole to be drilled first, while a hole with a large diameter might be machined with an endmill instead of a drill. By spanning a large range of dimensions and tool types, our dataset aims to capture this diversity of operations; identifying an operation only as drilling or milling is not sufficient to capture such variety.

Finally, the dataset provides several complementary modalities and annotations. We include images of the CAD model in wireframe and shaded views, which can be used for feature recognition and classification tasks; since modern deep learning techniques can ingest multiple modalities to make better predictions, these images provide an additional input channel alongside the geometry. We also report machining time estimates for each operation, with cutting and non-cutting times listed separately, together with the number of tool changes per part. These can be used to train models to predict machining time and tool changes, which are important metrics for process planning, as well as models for process optimization and scheduling. Lastly, we provide the tool paths for each operation, which can be used to train models to predict tool paths and optimize them for efficiency and accuracy.

3 Comparison to Other Datasets

Most large public CAD datasets target geometry or design intent rather than manufacturing. ABC [6] offers one million BRep models harvested from Onshape, but they carry no manufacturing labels and many are not machinable. The Fusion 360 Gallery [9] provides human design sequences with face-level labels, yet those labels describe CAD construction operations (extrude, fillet, chamfer) rather than machining operations. DeepCAD [10] similarly captures sketch-and-extrude command sequences for generative modeling. MFCAD [3] and MFCAD++ [4] are the closest manufacturing-oriented datasets, with 15,488 and 59,655 synthetic BRep models labeled per face with machining features, but they stop at feature recognition and contain no operation sequence, tool assignment, IPW state, machining time, or tool path. MachinePlan-10K instead labels the full process plan: for each part it records the ordered operations, the specific tool from the NX CAM tool library, the simulated IPW after every operation, cutting and non-cutting times, and the generated tool paths. It is thus complementary to the datasets above: those datasets emphasize geometric diversity, whereas we focus on the machining process, the sequence of operations, and the intermediate geometries.

DeepMS [7] is the only prior work that attempted to predict IPW and machining sequence using deep learning. Towards that, DeepMS provides a dataset for machining process planning similar to the one described in this paper. Table 1 presents a detailed comparison between the two datasets. The main differences are that we leverage BRep files and STL files for part representation, instead of voxels. That choice enables us to model parts and features at a higher resolution. We generate features first and then valid sequences using the machining knowledge base within NX CAM. We account for the availability and limitations of tools in the tool library. Additionally, we consider more variations of features such as hole patterns and through/blind holes, and we account for tool characteristics like flute length and corner radius constraints. We also simulate the machining process to generate the IPW, which is more realistic than CAD subtraction. Finally, we provide additional information such as part images, machining time estimates, tool paths, and tool types and sizes. These additional features make the MachinePlan-10K dataset more comprehensive and increase the options for training and evaluating modern deep learning models for machining process planning.

Table 1. Comparison between DeepMS and MachinePlan-10K.

Aspect	DeepMS	MachinePlan-10K
Count of parts	180,000 unique parts (1.08M augmented parts)	10,000 unique parts
Dimensions	100–128 voxels length, 100–128 voxels width, 50–80 voxels height	200–500 mm length, 200–500 mm width, 50–150 mm height
Resolution	1 voxel	0.01 mm
Process plan simulation	No	Yes
IPW generation strategy	CAD subtraction	Machining simulation
Part representation	Voxels	BRep files & STL files
Generation approach	Valid sequences → features	Features → valid sequences
Operation labels	Limited (drilling, milling, slant)	Extensive (3 main labels with 7 sub-labels)
Tool library integration	No	Yes
Flute length consideration	No	Yes
Corner radius consideration	No	Yes
Feature variations	Limited	Extensive (patterns, through/blind holes)
Machining knowledge base	No	Yes
Part images	No	Yes (wireframe and shaded)
Machining time estimates	No	Yes
Tool paths	No	Yes
Tool types and sizes	No	Yes

4 Dataset Structure

The dataset is structured into folders, with each folder containing the CAD model, CAM operations, tool paths, and images for a specific part. Each part is stored in its own subfolder at the dataset root, named after the part (e.g., `part_00001`). Within a part folder, part-level files describe the raw stock and the overall CAM operation sequence, while the machining operations themselves are stored as a numbered, zero-padded sequence of files. Each operation is identified by its index, the tool used (e.g., `UGT0301_287`), and a file suffix indicating its content: `_details.txt` contains the NX operation and tool report, `_text.stl.txt` is the IPW mesh after the operation, and `.ptp` is the generated tool path file. Figure 1 shows the representative folder structure for a single part. The four rendered images stored with every part are illustrated in Figure 2, where the three wireframe views are laid out in third-angle projection alongside the shaded isometric view.

5 Data Generation Pipeline

The data were generated using the modeling and manufacturing modules within NX (version 2512). The generation was automated using the NX Open API provided with the NX installation and Python (version 3.12) [8]. In Sections 6

```
MachinePlan-10K/
|-- part_00001/
|   |-- part_00001.stp                    % STEP export of CAD model
|   |-- part_00001_operations.json        % CAM operation sequence metadata
|   |-- front_wireframe.png               % front view wireframe image of CAD model
|   |-- top_wireframe.png                 % top view wireframe image of CAD model
|   |-- right_wireframe.png               % right view wireframe image of CAD model
|   |-- isometric_shaded.png              % shaded image of CAD model
|   |-- 000_BLANK_text.stl.txt            % IPW mesh before first operation
|   |-- 001_UGT0205_001_details.txt       % operation 1: tool/operation report
|   |-- 001_UGT0205_001_text.stl.txt      % operation 1: IPW mesh
|   |-- 001_UGT0205_001.ptp               % operation 1: tool path
|   |-- 002_UGT0205_001_details.txt
|   |-- 002_UGT0205_001_text.stl.txt
|   |-- 002_UGT0205_001.ptp
|   |-- ...
|   |-- 007_UGT0301_287_details.txt       % operation N: tool/operation report
|   |-- 007_UGT0301_287_text.stl.txt      % operation N: IPW mesh
|   `-- 007_UGT0301_287.ptp               % operation N: tool path
|-- part_00002/
|   `-- ...
`-- ...
```

Fig. 1. Representative folder structure of a single generated part. Every part subfolder follows the same layout, with the number of numbered operation triplets varying by part.

and 7, we describe the CAD and CAM data generation process in detail. The overall data generation pipeline is shown in Figure 3, with CAD model generation preceding CAM operations.

6 Using CAD for Generating the Part Model

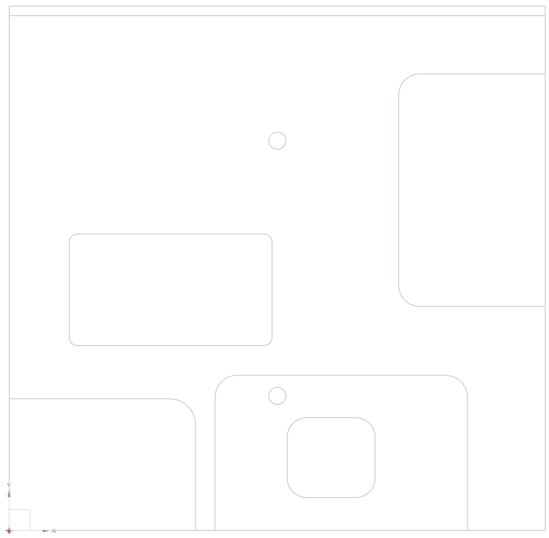
CAD model creation involves defining the raw stock, adding features, and ensuring that all features are valid and do not violate any geometric constraints.

6.1 Components of the CAD Model of a Part

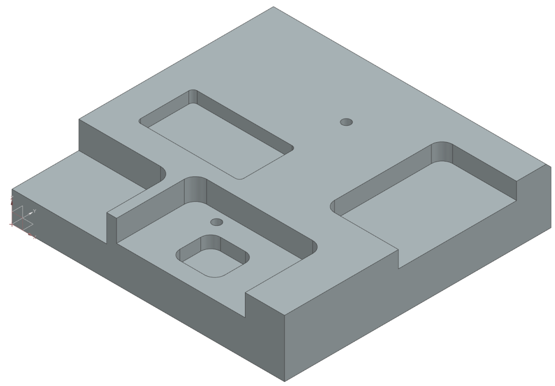
6.1.1 Raw stock. The raw stock is a cuboidal block created using the NX Open API, which serves as the starting point for the CAD models.

6.1.2 Features. We consider three types of features: pockets, holes, and chamfers. Features are created using the NX Open API and added to the raw stock. They are further classified based on their location, dimensions, and relationships with other features. Holes can be either through-holes or blind holes, and may form a pattern of identical holes or be placed as standalone instances. Pockets can be a corner pocket, edge pocket, channel/slot, or center pocket. A chamfer can be on any of the four top edges of the raw stock. Features may also interact: a hole can be nested inside a pocket, a chamfer can intersect a pocket or a hole, a pocket can be nested within another pocket, a pocket can intersect a hole, and a pocket can abut another pocket. These interactions provide a rich space of features and feature combinations.

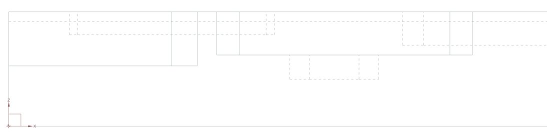
6.1.3 Feature parameters. Blind holes are parameterized by their depth, diameter, and position. Through-holes are parameterized by their diameter and position. Hole patterns are parameterized by the number of holes and the pattern



(a) Top view (XY), wireframe.



(b) Isometric view, shaded.



(c) Front view (XZ), wireframe.

(d) Right view (YZ), wireframe.

Fig. 2. The four rendered images stored in every part folder, shown here for `part_00001`. The three orthographic wireframe views follow a third-angle arrangement: the top view (a) sits above the front view (c), and the right view (d) is placed beside the front view, so the same three coordinate directions are read across and down the page. Dashed lines denote hidden edges, keeping pockets visible from every direction. The shaded isometric view (b) occupies the remaining quadrant and conveys the overall three-dimensional shape. Each subfigure has been cropped to its part geometry, so the views are not drawn to a common scale.

type (linear, circular, grid, etc.). Pockets are parameterized by their depth, width, length, position, and corner blend radius. Chamfers are parameterized by their width and edge location.

6.1.4 Valid parameter ranges. The valid range of parameters depends on the already existing features on the block. The range also depends on the size of the block itself. Holes are designed such that there is at least a pre-defined gap between the hole and the edge of the block. The parameter ranges for each feature type also depend on the tools available in the NX CAM tool library. The parameter ranges are defined to ensure that the features can be machined using the available tools.

6.2 Feature Sampling Scheme

The number of features of each type, their parameters, and their relationships are chosen by a random draw for each part. A fixed random seed is used to ensure reproducibility of the dataset. A summary of the sampling parameters is shown in Table 2. The seed for the first part is 12345, incremented by 1 for each subsequent part. Each part is generated by sampling the raw block and then adding features in a fixed order: chamfers first, followed by pockets, and finally holes. A coordinate frame is defined with the block corner at the origin and the top face at $z = H$, so that every feature

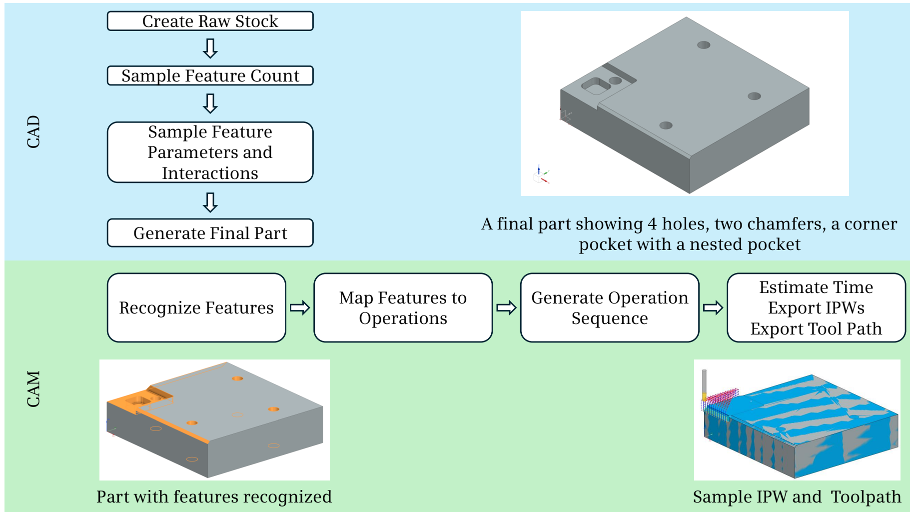


Fig. 3. Overall data generation pipeline. The CAD models are generated first, followed by the CAM operations. The CAM operations are then used to generate the tool paths and IPWs.

Table 2. Default sampling parameters used for data generation.

Parameter	Value
Block length L	$U(200, 500)$ mm
Block width W	$U(200, 500)$ mm
Block height H	$U(50, 150)$ mm
Number of chamfers (0–4)	weights 0.15, 0.35, 0.30, 0.15, 0.05
Number of pockets (0–5)	weights 0.15, 0.30, 0.25, 0.15, 0.10, 0.05
Number of holes (0–6)	weights 0.15, 0.20, 0.20, 0.15, 0.15, 0.10, 0.05
Minimum chamfer distance	2.0 mm
Pocket type	corner 0.25, edge 0.25, center 0.30, slot 0.20
Pocket relation (2nd+ pocket)	standalone 0.50, abutting 0.25, nested 0.25
Pocket depth fraction of H	$U(0.10, 0.60)$
Nested pocket extra depth fraction of H	$U(0.10, 0.40)$
Pocket fillet radius	$U(3.0, 15.0)$ mm
P(pocket–chamfer intersection)	0.30
P(hole pattern vs. random)	0.40
Hole pattern type	linear 0.40, grid 0.30, circular 0.30
P(hole through vs. blind)	0.50
Hole diameter	5.0–50.0 mm
Blind hole depth fraction of H	$U(0.20, 0.70)$
P(hole–pocket intersection)	0.30
P(hole–chamfer intersection)	0.20
Edge margin	4.0 mm
Max resampling retries	25

is cut downward ($-Z$) from the top face. The block length L, width W, and height H are each drawn independently from their respective uniform ranges.

- **Chamfers.** The number of chamfers is drawn from a categorical distribution over $\{0, \ldots, 4\}$ with weights $0.15, 0.35, 0.30, 0.15, 0.05$, and that many distinct top edges are chosen at random from the four candidates (x_0, x_1, y_0, y_1). Each chamfer distance is then drawn uniformly between a minimum of 2.0 mm and a maximum of $\min(0.2 \times \min(L, W), 0.2H)$.
- **Pockets.** The number of pockets is drawn from a categorical distribution over $\{0, \ldots, 5\}$ with weights $0.15, 0.30, 0.25, 0.15, 0.10, 0.05$. Each pocket is assigned a type of corner, edge, center, or slot with probabilities 0.25, 0.25, 0.30, and 0.20 respectively, where a slot is a through-channel spanning the full block length or width, open at both ends with two side walls and a floor; slots are always standalone and never nested, abutting, or filleted. Starting with the second pocket, a relation to the previously placed pockets is sampled as standalone, abutting, or nested with probabilities 0.50, 0.25, and 0.25. A fillet radius on the interior vertical corner edges is drawn uniformly from $U(3.0, 15.0)$ mm, capped to fit within the pocket footprint (no more than 0.45 times the smaller of the pocket length and width); slots have no interior vertical corners and therefore no fillet. Pocket depth is sampled as a fraction $U(0.10, 0.60)$ of H (with an additional $U(0.10, 0.40) \times H$ for the extra depth of a nested pocket), then clamped to the flute-length feasibility cap described below. Finally, whether the pocket intersects a chamfer is determined by a Bernoulli draw with probability 0.30.
- **Holes.** The number of holes is drawn from a categorical distribution over $\{0, \ldots, 6\}$ with weights $0.15, 0.20, 0.20, 0.15, 0.15, 0.10, 0.05$. A Bernoulli draw with probability 0.40 decides whether the holes are arranged in a shared pattern (linear, grid, or circular, with probabilities 0.40, 0.30, and 0.30) with a common diameter and depth type, or otherwise placed randomly with independent centers. Because holes are drilled, each diameter is matched to the nearest discrete drill size from a tabulated flute-length reference based on tools in NX CAM, and restricted to the range 5.0–50.0 mm and to sizes with an available flute length. A further Bernoulli draw with probability 0.50 determines whether a hole is through or blind: through-holes cut the entire height H and may only use drill sizes whose flute length exceeds H, so the tool can reach through (if no such size exists, the hole is instead made blind); blind holes have a depth sampled as a fraction $U(0.20, 0.70)$ of H and capped strictly below the drill's maximum flute length. Whether a hole intersects a pocket or a chamfer is likewise determined by independent Bernoulli draws with probabilities 0.30 and 0.20 respectively.

Tool feasibility limits the achievable depth of a pocket: an interior vertical corner of radius r can only be machined by an endmill of diameter no greater than $2r$, and the flute length of the deepest such tool sets the maximum reachable depth from the top face. This cap is obtained from the endmill flute-length table as the maximum flute length among tabulated diameters not exceeding $2r$, and every sampled pocket depth is clamped to it. For a nested pocket, the total depth from the top face (the parent pocket's depth plus the additional nested depth) is clamped to the cap derived from the nested pocket's own, smaller fillet radius.

More generally, every feature footprint must remain inside the block boundary $[0, L] \times [0, W]$ with a margin of 4.0 mm from each edge for holes. All sizes are clamped to the configured minimum and maximum tool ranges. Overlap rules are enforced according to the sampled relation: standalone features may not overlap, abutting features may only touch, and nested features must be fully contained within their parent. If a sampled draw is infeasible, it is re-sampled up to 25 retries; if it remains infeasible, the feature is skipped. Because the achieved relations and interactions are recomputed directly from the final geometry and written to the output metadata, the resulting JSON always accurately reflects the generated part.

6.3 Verification of the CAD Models

NX has built-in geometry checks that validate the CAD models and report errors if any issues are detected. Beyond these, geometric checks are performed using Python scripts to validate feature parameters and geometry. The checks include:

- Ensuring that holes do not overlap with each other.
- A hole can only be nested inside a pocket if it is deeper than the pocket.
- Ensuring that pockets do not overlap with each other unless they are abutting or nested.
- Verifying that holes or pockets do not intersect with chamfers, unless the intersection is allowed by the sampled parameters.
- Checking that pockets do not exceed the boundaries of the raw stock.
- Validating that all feature parameters are within the defined ranges.

6.4 Exporting BRep (.stp) Files and Images

Once the CAD model is created, it is exported as a BRep file in STEP (`.stp`) format. We also capture images of the CAD models using the NX Open API. We create wireframe images according to third-angle projection as per ASME guidelines [1]. A shaded isometric view of the part is also created.

6.5 Robustness of CAD Generation

Within the CAD module, no complete part-generation failures occurred. Individual feature placements that remained infeasible after 25 retries were skipped and replaced by other valid feature configurations. However, generating a CAD model does not guarantee that it can be machined successfully. The CAM generation process is more complex and may fail for some parts: operations may be missing for certain features due to tool constraints, and an assigned operation may not remove 100% of the required material. More than 10,000 parts were therefore generated to account for such failures or incomplete machining, and any parts that fell below the acceptance thresholds (described in Section 7.2) were discarded. The CAM generation process is described in Section 7.

6.6 Summary of the Generated Parts and Features

In total, 11,416 parts were generated by the CAD pipeline before reaching 10,000 parts that met the acceptance criteria. A summary across the 10,000 retained parts is provided below, covering feature prevalence, part size, and machining complexity.

- Grand total features generated: 58,086
- 16,280 chamfers, 18,484 pockets, 23,322 holes
- Total features per part (chamfers + pockets + holes): mean 5.81, median 6.0, min 1.0, max 14.0, std. dev. 2.40
- Parts with at least one pocket: 8,540 (85.4%)
- Parts with at least one hole: 8,357 (83.6%)
- Parts with at least one chamfer: 8,566 (85.7%)
- Parts with all three feature types: 6,052 (60.5%)
- Operations per part: mean 9.17, min 1.0, max 38.0, std. dev. 5.21

At the part level, the mean counts per part were 2.33 holes (median 2.0, range 0–6), 1.85 pockets (median 2.0, range 0–5), and 1.63 chamfers (median 2.0, range 0–4). Feature combination frequencies are as follows: 211 parts (2.1%) had pockets only, 146 (1.5%) had holes only, 232 (2.3%) had chamfers only, 1,077 (10.8%) had pockets and holes, 1,200 (12.0%) had pockets and chamfers, 1,082 (10.8%) had holes and chamfers, and 6,052 (60.5%) had all three feature types.

As for the distribution of the part size (bounding box, mm):

- Length: mean 350.22 mm, min 200.0 mm, max 500.0 mm, std. dev. 85.84 mm
- Width: mean 350.61 mm, min 200.0 mm, max 500.0 mm, std. dev. 86.11 mm
- Height: mean 100.05 mm, min 50.0 mm, max 150.0 mm, std. dev. 29.11 mm
- Bounding-box volume: mean 12,252,896 mm^3, min 2,246,315 mm^3, max 35,467,801 mm^3, std. dev. 5,677,696 mm^3

Feature dimensions were broadly distributed across the configured ranges. Hole diameters ranged from 5.0 to 50.0 mm (mean 17.47 mm), while hole depths ranged from 10.1 to 150.0 mm (mean 71.99 mm); through-holes had mean depth 99.36 mm and blind holes had mean depth 44.89 mm. Pocket width and length were similar on average (means 106.40 mm and 107.24 mm), with depth mean 28.94 mm and fillet radius mean 7.63 mm; pocket footprint area ranged from 204.49 to 42,348.80 mm^2 (mean 8,891.35 mm^2). Chamfer distance ranged from 2.0 to 29.7 mm with mean 11.05 mm.

The generated frequency of feature categories and interactions is summarized below:

- Hole depth types (23,322 total holes): 11,720 blind (50.2%) and 11,602 through (49.8%).
- Pocket types (18,484 total pockets): 5,329 center (28.8%), 5,155 corner (27.9%), 5,131 edge (27.8%), and 2,869 slot (15.5%).
- Pocket relations: 15,797 standalone (85.5%), 2,442 nested (13.2%), and 245 abutting (1.3%).
- Chamfer edge distribution (16,280 total chamfers): nearly balanced across x_0 (4,075), x_1 (4,039), y_0 (4,040), and y_1 (4,126).
- Pocket–chamfer interactions: 7,990 total interactions, with 6,514 pockets touching at least one chamfer.
- Hole–pocket interactions: 5,506 holes touched at least one pocket, giving 5,919 hole–pocket pairs (a hole intersecting more than one pocket contributes one pair per pocket). Pair counts were blind–center (986), blind–corner (708), blind–edge (825), blind–slot (506), through–center (895), through–corner (665), through–edge (773), and through–slot (561).
- Hole–chamfer interactions: 3,238 holes touched at least one chamfer.

7 Using CAM for Generating Machining Operations

Once the CAD models have been created, the CAM module is used to generate the machining operations. The operations are generated based on the features present in the CAD model. They are then sequenced according to feature interactions and the machining rules defined in the NX knowledge library. Finally, tool paths and IPWs are generated for each operation. The following steps are followed to generate the CAM operations:

(1) *Feature Recognition*: Feature recognition is performed using the NX Open API. Features are identified from part geometry using the knowledge-base rules within NX.
(2) *Feature to Operation Mapping*: Features are then mapped to operations based on their type and dimensions.
(3) *Operation to Tool Mapping*: Operations are then mapped to tools based on the operation type and the dimensions of the feature, along with the tool library available in NX.
(4) *Tool Path and Operation Sequence Generation*: Once operations and tools are mapped, the tool path is generated using NX CAM. The operation sequence is generated based on the rules defined in the knowledge base within NX.

(5) *Time Estimate*: NX CAM provides time estimates for each operation based on the tool path generated. The time estimates are based on the feed rates, spindle speeds, and other parameters defined in the tool library.
(6) *IPW Export*: The IPW is exported from NX CAM after each operation. The IPW contains the geometry of the part after each operation. IPWs can be used for verification and validation of the machining process.

7.1 Operation Label Descriptions

For each operation, the operation type, subtype, tool type, tool diameter, and tool length are recorded in the operation details file. The spindle speed, feed rate, and other parameters are also recorded along with the NC code in the `.ptp` files.

The dataset contains three high-level operation types: hole-making, contour milling, and planar milling. Table 3 shows the distribution across the 91,702 total operations, with hole-making operations accounting for the majority owing to the prevalence of holes and hole patterns in the generated parts.

Table 3. Operation type labels and frequencies.

Operation Type	Count
hole_making	53,075
mill_contour	20,067
mill_planar	18,560

Each operation type is further subdivided into more specific subtypes that reflect the machining strategy employed. Table 4 lists all seven subtypes and their frequencies. DRILLING and AREA_MILL are the most common, corresponding to the high prevalence of drilled holes and chamfer milling operations, respectively.

Table 4. Operation subtype labels and frequencies.

Operation Subtype	Count
DRILLING	30,377
AREA_MILL	20,067
FLOOR_WALL	18,560
SPOT_DRILLING	14,189
DEEP_HOLE_DRILLING	4,467
HOLE_MILLING	3,470
BORING_REAMING	572

Table 5 lists the ten most frequently selected tools across all operations. The chamfer mill dominates due to its use in every chamfer milling operation, while spot drills and endmills of various diameters reflect the range of hole and pocket dimensions present in the dataset.

7.2 Robustness of CAM Generation

The NX CAM module flags the most common errors during the generation of machining operations. Beyond that, every generated part was evaluated using multiple tests to ensure that the CAM operations were generated correctly. The tests included:

Table 5. Ten most frequently used tools in the dataset.

Tool name	Count
Chamfer Mill 20 x 3 x 45 deg.	20,067
NC Spot Drill, 142 deg, 12 mm, Carbide, Uncoated	14,150
ENDMILL 16 mm (2P460-1600-NA 1630)	3,372
End Mill 20 mm	3,073
ENDMILL 12 mm (2P460-1200-NA 1630)	2,925
ENDMILL 10 mm	1,796
Carbide Drill 9.4 mm (860.1-0940-075A1-MM 2214)	1,536
End Mill 16 mm	1,119
ENDMILL 8 mm (2P460-0800-NA 1630)	1,119
Carbide Drill 12 mm (861.1-1200-036A1-GP GC34)	892

- The percentage of material removed by the operations, to verify that the correct amount of material has been removed. We only considered parts where the percentage of material removed was greater than 99%.
- The Intersection over Union (IoU) between the final IPW and the CAD model, to verify that the final part geometry matches the target CAD model. For the parts with more than 99% material removed, we further filtered to consider parts with an IoU greater than 0.999.

After these tests, 10,000 parts were found to be valid among the first 11,416 parts generated. This translates to a success rate of approximately 87.6%. The remaining 1,416 parts were discarded due to errors in the CAM generation process or insufficient volume removal. This is expected, as the NX CAM module may fail to generate valid operations for certain parts. Note also that 100% material removal is not always possible in subtractive manufacturing, and minor mismatches due to tool availability are unavoidable.

8 Limitations

Some limitations of the dataset are as follows:

- The dataset is limited to planar machining operations under 2.5-axis CNC machining.
- The dataset does not include complex features such as threaded holes, countersunk holes, or counterbored holes.
- The dataset is entirely synthetic and does not account for real-world variations in material properties, machine behavior, and other external factors.
- The dataset does not include multiple valid sequences for a given part; only one valid sequence is provided per part.
- The dataset assumes that the tool library is fixed and does not account for the availability of different tools in different machines.

9 Possible Extensions

Some possible extensions to the dataset include:

- Additional features such as bosses and chamfers with different angles.
- Additional operations such as threading, countersinking, and counterboring.
- Multiple valid machining sequences for a given part.

10 Conclusion

The dataset described in this paper provides a comprehensive set of machining scenarios for planar machining operations under 2.5-axis CNC machining. The dataset includes a variety of features, operations, and tool paths, along with images and time estimates for each operation. It can be used for research and development in machining process planning, and extended to incorporate additional features, operations, and tool paths as needed.

11 Data Availability

The dataset is available for download at https://doi.org/10.5281/zenodo.21653081 [2]. The dataset is provided under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which allows for sharing and adaptation with appropriate credit.

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